QUILLEN SPECTRAL SEQUENCE FROM SYMMETRIC POWERS OF COTANGENT DATA TO TOR

THE STACKS PROJECT AUTHORS

Editorial note. The09CG task identifies a three-term truncation of the cotangent complex. This task constructs an all-degree spectral sequence and proves its convergence; neither statement is a relabelled proof of the other.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Simplicial polynomial resolutions, derived lower shriek, filtered-complex spectral sequences, change-of-ringsTor and truncation composition. The local comparison, symmetric-power vanishing and Tor-map-zero proofs are retained.

Source reference. Original lemma 08RX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11110–11110. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08QI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11061–11061. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08PF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11023–11023. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08Q9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11052–11052. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08QC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11055–11055. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00RP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1011–1011. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:cotangent.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 08Q8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11051–11051. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 012K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1402–1402. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08Q2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11045–11045. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 068F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8004–8004. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08N8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10981–10981. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 09CB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11824–11824. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0194: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1639–1639. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 031G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3983–3983. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 07GE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9514–9514. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08NN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10996–10996. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08NR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10999–10999. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08NS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11000–11000. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08ND: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10986–10986. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08P2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11010–11010. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08P1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11009–11009. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00RR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1013–1013. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1016–1016. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Theorem 0.1 (Quillen spectral sequence)**.** *Let $A \to B$ be a surjective ring map. Consider the sheaf $\Omega = \Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B}$ of $\underline{B}$-modules on $\mathcal{C}_{B/A}$, see Section 2. Then there is a spectral sequence with E_1-page*

$$E_1^{p,q} = H_{-p-q}(\mathcal{C}_{B/A}, Sym^p_{\underline{B}}(\Omega)) \Rightarrow Tor^A_{-p-q}(B, B)$$

with d_r of bidegree $(r, -r+1)$. Moreover, $H_i(\mathcal{C}_{B/A}, Sym^k_{\underline{B}}(\Omega)) = 0$ for $i < k$.

1. The cotangent complex of a ring map

Let A be a ring. Let Alg_A be the category of A-algebras. Consider the pair of adjoint functors (U, V) where $V : Alg_A \to Sets$ is the forgetful functor and $U : Sets \to Alg_A$ assigns to a set E the polynomial algebra $A[E]$ on E over A. Let $X_\bullet$ be the simplicial object of $\mathrm{Fun}(Alg_A, Alg_A)$ constructed in Simplicial, Section 08N8.

Consider an A-algebra B. Denote $P_\bullet = X_\bullet(B)$ the resulting simplicial A-algebra. Recall that $P_0 = A[B]$, $P_1 = A[A[B]]$, and so on. In particular each term P_n is a polynomial A-algebra. Recall also that there is an augmentation

$$\epsilon : P_\bullet \longrightarrow B$$

where we view B as a constant simplicial A-algebra.

Definition 1.1. Let $A \to B$ be a ring map. The *standard resolution of B over A* is the augmentation $\epsilon : P_\bullet \to B$ with terms

$$P_0 = A[B], \quad P_1 = A[A[B]], \quad \ldots$$

and maps as constructed in Simplicial, Example 09CB.

It will turn out that we can use the standard resolution to compute left derived functors in certain settings.

Definition 1.2. The *cotangent complex* $L_{B/A}$ of a ring map $A \to B$ is the complex of B-modules associated to the simplicial B-module

$$\Omega_{P_\bullet/A} \otimes_{P_\bullet, \epsilon} B$$

where $\epsilon : P_\bullet \to B$ is the standard resolution of B over A.

In Simplicial, Section 0194 we associate a chain complex to a simplicial module, but here we work with cochain complexes. Thus the term $L^{-n}_{B/A}$ in degree $-n$ is the B-module $\Omega_{P_n/A} \otimes_{P_n, \epsilon_n} B$ and $L^m_{B/A} = 0$ for $m > 0$.

Remark 1.3. Let $A \to B$ be a ring map. Let $\mathcal{A}$ be the category of arrows $\psi : C \to B$ of A-algebras and let $\mathcal{S}$ be the category of maps $E \to B$ where E is a set. There are adjoint functors $V : \mathcal{A} \to \mathcal{S}$ (the forgetful functor) and $U : \mathcal{S} \to \mathcal{A}$ which sends $E \to B$ to $A[E] \to B$. Let $X_\bullet$ be the simplicial object of $\mathrm{Fun}(\mathcal{A}, \mathcal{A})$ constructed in Simplicial, Section 08N8. The diagram

$$\begin{array}{ccc} \mathcal{A} & \rightleftarrows & \mathcal{S} \\ \downarrow & & \downarrow \\ Alg_A & \rightleftarrows & Sets \end{array}$$

commutes. It follows that $X_\bullet(\mathrm{id}_B : B \to B)$ is equal to the standard resolution of B over A.

Lemma 1.4. *Let $A_i \to B_i$ be a system of ring maps over a directed index set I. Then* $\operatorname{colim} L_{B_i/A_i} = L_{\operatorname{colim} B_i / \operatorname{colim} A_i}$.

Proof. This is true because the forgetful functor $V : A\text{-}Alg \to Sets$ and its adjoint $U : Sets \to A\text{-}Alg$ commute with filtered colimits. Moreover, the functor $B/A \mapsto \Omega_{B/A}$ does as well (Algebra, Lemma 031G). □

2. Simplicial resolutions and derived lower shriek

Let $A \to B$ be a ring map. Consider the category whose objects are A-algebra maps $\alpha : P \to B$ where P is a polynomial algebra over A (in some set[1] of variables) and whose morphisms $s : (\alpha : P \to B) \to (\alpha' : P' \to B)$ are A-algebra homomorphisms $s : P \to P'$ with $\alpha' \circ s = \alpha$. Let $\mathcal{C} = \mathcal{C}_{B/A}$ denote the **opposite** of this category.

[1] It suffices to consider sets of cardinality at most the cardinality of B.

The reason for taking the opposite is that we want to think of objects (P, α) as corresponding to the diagram of affine schemes

$$\begin{array}{ccc} \mathrm{Spec}(B) & \longrightarrow & \mathrm{Spec}(P) \\ & \searrow & \downarrow \\ & & \mathrm{Spec}(A) \end{array}$$

We endow $\mathcal{C}$ with the chaotic topology (Sites, Example 07GE), i.e., we endow $\mathcal{C}$ with the structure of a site where coverings are given by identities so that all presheaves are sheaves. Moreover, we endow $\mathcal{C}$ with two sheaves of rings. The first is the sheaf $\mathcal{O}$ which sends to object (P, α) to P. Then second is the constant sheaf B, which we will denote $\underline{B}$. We obtain the following diagram of morphisms of ringed topoi

$$\begin{array}{ccc} (Sh(\mathcal{C}), \underline{B}) & \xrightarrow[i]{} & (Sh(\mathcal{C}), \mathcal{O}) \\ \downarrow{\scriptstyle \pi} & & \\ (Sh(*), B) & & \end{array} \tag{2.0.1}$$

The morphism i is the identity on underlying topoi and $i^\sharp : \mathcal{O} \to \underline{B}$ is the obvious map. The map π is as in Cohomology on Sites, Example 08PF. An important role will be played in the following by the derived functors $Li^* : D(\mathcal{O}) \longrightarrow D(\underline{B})$ left adjoint to $Ri_* = i_* : D(\underline{B}) \to D(\mathcal{O})$ and $L\pi_! : D(\underline{B}) \longrightarrow D(B)$ left adjoint to $\pi^* = \pi^{-1} : D(B) \to D(\underline{B})$.

Lemma 2.1. *With notation as above let $P_\bullet$ be a simplicial A-algebra endowed with an augmentation $\epsilon : P_\bullet \to B$. Assume each P_n is a polynomial algebra over A and ϵ is a trivial Kan fibration on underlying simplicial sets. Then*

$$L\pi_!(\mathcal{F}) = \mathcal{F}(P_\bullet, \epsilon)$$

in $D(Ab)$, resp. $D(B)$ functorially in $\mathcal{F}$ in $Ab(\mathcal{C})$, resp. $Mod(\underline{B})$.

Proof. We will use the criterion of Cohomology on Sites, Lemma 08Q9 to prove this. Given an object $U = (Q, \beta)$ of $\mathcal{C}$ we have to show that

$$S_\bullet = \mathrm{Mor}_{\mathcal{C}}((P_\bullet, \epsilon), (Q, \beta))$$

is homotopy equivalent to a singleton. Write $Q = A[E]$ for some set E (this is possible by our choice of the category $\mathcal{C}$). We see that

$$S_\bullet = \mathrm{Mor}_{Sets}((E, \beta|_E), (P_\bullet, \epsilon))$$

Let $*$ be the constant simplicial set on a singleton. For $b \in B$ let $F_{b,\bullet}$ be the simplicial set defined by the cartesian diagram

$$\begin{array}{ccc} F_{b,\bullet} & \longrightarrow & P_\bullet \\ \downarrow & & \downarrow{\scriptstyle \epsilon} \\ * & \xrightarrow{b} & B \end{array}$$

With this notation $S_\bullet = \prod_{e \in E} F_{\beta(e),\bullet}$. Since we assumed ϵ is a trivial Kan fibration we see that $F_{b,\bullet} \to *$ is a trivial Kan fibration (Simplicial, Lemma 08NN). Thus $S_\bullet \to *$ is a trivial Kan fibration (Simplicial, Lemma 08NR). Therefore $S_\bullet$ is homotopy equivalent to $*$ (Simplicial, Lemma 08NS). □

In particular, we can use the standard resolution of B over A to compute derived lower shriek.

Lemma 2.2. *Let $A \to B$ be a ring map. Let $\epsilon : P_\bullet \to B$ be the standard resolution of B over A. Let π be as in (2.0.1). Then*

$$L\pi_!(\mathcal{F}) = \mathcal{F}(P_\bullet, \epsilon)$$

in $D(Ab)$, resp. $D(B)$ functorially in $\mathcal{F}$ in $Ab(\mathcal{C})$, resp. $Mod(\underline{B})$.

First proof. We will apply Lemma 2.1. Since the terms P_n are polynomial algebras we see the first assumption of that lemma is satisfied. The second assumption is proved as follows. By Simplicial, Lemma 08ND the map ϵ is a homotopy equivalence of underlying simplicial sets. By Simplicial, Lemma 08P2 this implies ϵ induces a quasi-isomorphism of associated complexes of abelian groups. By Simplicial, Lemma 08P1 this implies that ϵ is a trivial Kan fibration of underlying simplicial sets. □

Second proof. We will use the criterion of Cohomology on Sites, Lemma 08Q9. Let $U = (Q, \beta)$ be an object of $\mathcal{C}$. We have to show that

$$S_\bullet = \mathrm{Mor}_{\mathcal{C}}((P_\bullet, \epsilon), (Q, \beta))$$

is homotopy equivalent to a singleton. Write $Q = A[E]$ for some set E (this is possible by our choice of the category $\mathcal{C}$). Using the notation of Remark 1.3 we see that

$$S_\bullet = \mathrm{Mor}_{\mathcal{S}}((E \to B), i(P_\bullet \to B))$$

By Simplicial, Lemma 08ND the map $i(P_\bullet \to B) \to i(B \to B)$ is a homotopy equivalence in $\mathcal{S}$. Hence $S_\bullet$ is homotopy equivalent to

$$\mathrm{Mor}_{\mathcal{S}}((E \to B), (B \to B)) = \{*\}$$

as desired. □

Lemma 2.3. *Let $A \to B$ be a ring map. Let π and i be as in (2.0.1). There is a canonical isomorphism*

$$L_{B/A} = L\pi_!(Li^*\Omega_{\mathcal{O}/A}) = L\pi_!(i^*\Omega_{\mathcal{O}/A}) = L\pi_!(\Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B})$$

in $D(B)$.

Proof. For an object $\alpha : P \to B$ of the category $\mathcal{C}$ the module $\Omega_{P/A}$ is a free P-module. Thus $\Omega_{\mathcal{O}/A}$ is a flat $\mathcal{O}$-module. Hence $Li^*\Omega_{\mathcal{O}/A} = i^*\Omega_{\mathcal{O}/A}$ is the sheaf of $\underline{B}$-modules which associates to $\alpha : P \to A$ the B-module $\Omega_{P/A} \otimes_{P,\alpha} B$. By Lemma 2.2 we see that the right hand side is computed by the value of this sheaf on the standard resolution which is our definition of the left hand side (Definition 1.2). □

Lemma 2.4. *If $A \to B$ is a ring map, then $L\pi_!(\pi^{-1}M) = M$ with π as in (2.0.1).*

Proof. This follows from Lemma 2.1 which tells us $L\pi_!(\pi^{-1}M)$ is computed by $(\pi^{-1}M)(P_\bullet, \epsilon)$ which is the constant simplicial object on M. □

Lemma 2.5. *If $A \to B$ is a ring map, then $H^0(L_{B/A}) = \Omega_{B/A}$.*

Proof. We will prove this by a direct calculation. We will use the identification of Lemma 2.3. There is clearly a map from $\Omega_{\mathcal{O}/A} \otimes \underline{B}$ to the constant sheaf with value $\Omega_{B/A}$. Thus this map induces a map

$$H^0(L_{B/A}) = H^0(L\pi_!(\Omega_{\mathcal{O}/A} \otimes \underline{B})) = \pi_!(\Omega_{\mathcal{O}/A} \otimes \underline{B}) \to \Omega_{B/A}$$

By choosing an object $P \to B$ of $\mathcal{C}_{B/A}$ with $P \to B$ surjective we see that this map is surjective (by Algebra, Lemma 00RR). To show that it is injective, suppose that $P \to B$ is an object of $\mathcal{C}_{B/A}$ and that $\xi \in \Omega_{P/A} \otimes_P B$ is an element which maps to zero in $\Omega_{B/A}$. We first choose factorization $P \to P' \to B$ such that $P' \to B$ is surjective and P' is a polynomial algebra over A. We may replace P by P'. If $B = P/I$, then the kernel $\Omega_{P/A} \otimes_P B \to \Omega_{B/A}$ is the image of I/I^2 (Algebra,

Lemma 00RU). Say ξ is the image of $f \in I$. Then we consider the two maps $a, b : P' = P[x] \to P$, the first of which maps x to 0 and the second of which maps x to f (in both cases $P[x] \to B$ maps x to zero). We see that ξ and 0 are the image of $\mathrm{d}x \otimes 1$ in $\Omega_{P'/A} \otimes_{P'} B$. Thus ξ and 0 have the same image in the colimit (see Cohomology on Sites, Example 08PF) $\pi_!(\Omega_{\mathcal{O}/A} \otimes \underline{B})$ as desired. □

Lemma 2.6. *If B is a polynomial algebra over the ring A, then with π as in (2.0.1) we have that $\pi_!$ is exact and $\pi_!\mathcal{F} = \mathcal{F}(B \to B)$.*

Proof. This follows from Lemma 2.1 which tells us the constant simplicial algebra on B can be used to compute $L\pi_!$. □

Lemma 2.7. *If B is a polynomial algebra over the ring A, then $L_{B/A}$ is quasi-isomorphic to $\Omega_{B/A}[0]$.*

Proof. Immediate from Lemmas 2.3 and 2.6. □

Lemma 2.8. *Let $A \to B$ be a ring map. Let π, $\mathcal{O}$, $\underline{B}$ be as in (2.0.1). For any $\mathcal{O}$-module $\mathcal{F}$ we have*

$$L\pi_!(\mathcal{F}) = L\pi_!(Li^*\mathcal{F}) = L\pi_!(\mathcal{F} \otimes^{\mathbf{L}}_{\mathcal{O}} \underline{B})$$

in $D(Ab)$.

Proof. It suffices to verify the assumptions of Cohomology on Sites, Lemma 08RX hold for $\mathcal{O} \to \underline{B}$ on $\mathcal{C}_{B/A}$. We will use the results of Remark 08QI without further mention. Choose a resolution $P_\bullet$ of B over A to get a suitable cosimplicial object $U_\bullet$ of $\mathcal{C}_{B/A}$. Since $P_\bullet \to B$ induces a quasi-isomorphism on associated complexes of abelian groups we see that $L\pi_!\mathcal{O} = B$. On the other hand $L\pi_!\underline{B}$ is computed by $\underline{B}(U_\bullet) = B$. This verifies the second assumption of Cohomology on Sites, Lemma 08RX and we are done with the proof. □

Lemma 2.9. *Let $A \to B$ be a ring map. Let π, $\mathcal{O}$, $\underline{B}$ be as in (2.0.1). We have*

$$L\pi_!(\mathcal{O}) = L\pi_!(\underline{B}) = B \quad \text{and} \quad L_{B/A} = L\pi_!(\Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B}) = L\pi_!(\Omega_{\mathcal{O}/A})$$

in $D(Ab)$.

Proof. This is just an application of Lemma 2.8 (and the first equality on the right is Lemma 2.3). □

Lemma 2.10. *Notation and assumptions as in Cohomology on Sites, Example 08PF. Assume $\mathcal{C}$ has a cosimplicial object as in Cohomology on Sites, Lemma 08Q9. Let $\mathcal{F}$ be a flat $\underline{B}$-module such that $H_0(\mathcal{C}, \mathcal{F}) = 0$. Then $H_l(\mathcal{C}, Sym^k_{\underline{B}}(\mathcal{F})) = 0$ for $l < k$.*

Proof. We drop the subscript $\underline{B}$ from tensor products, wedge powers, and symmetric powers. We will prove the lemma by induction on k. The cases $k = 0, 1$ follow from the assumptions. If $k > 1$ consider the exact complex

$$\ldots \to \wedge^2\mathcal{F} \otimes \mathrm{Sym}^{k-2}\mathcal{F} \to \mathcal{F} \otimes \mathrm{Sym}^{k-1}\mathcal{F} \to \mathrm{Sym}^k\mathcal{F} \to 0$$

with differentials as in the Koszul complex. If we think of this as a resolution of $\mathrm{Sym}^k\mathcal{F}$, then this gives a first quadrant spectral sequence

$$E_1^{p,q} = H_p(\mathcal{C}, \wedge^{q+1}\mathcal{F} \otimes \mathrm{Sym}^{k-q-1}\mathcal{F}) \Rightarrow H_{p+q}(\mathcal{C}, \mathrm{Sym}^k(\mathcal{F}))$$

By Cohomology on Sites, Lemma 08QC we have

$$L\pi_!(\wedge^{q+1}\mathcal{F} \otimes \mathrm{Sym}^{k-q-1}\mathcal{F}) = L\pi_!(\wedge^{q+1}\mathcal{F}) \otimes^{\mathbf{L}}_B L\pi_!(\mathrm{Sym}^{k-q-1}\mathcal{F}))$$

It follows (from the construction of derived tensor products) that the induction hypothesis combined with the vanishing of $H_0(\mathcal{C}, \wedge^{q+1}(\mathcal{F})) = 0$ will prove what we

want. This is true because $\wedge^{q+1}(\mathcal{F})$ is a quotient of $\mathcal{F}^{\otimes q+1}$ and $H_0(\mathcal{C}, \mathcal{F}^{\otimes q+1})$ is a quotient of $H_0(\mathcal{C}, \mathcal{F})^{\otimes q+1}$ which is zero. □

Remark 2.11. In the situation of Lemma 2.10 one can show that $H_k(\mathcal{C}, \mathrm{Sym}^k(\mathcal{F})) = \wedge^k_B(H_1(\mathcal{C}, \mathcal{F}))$. Namely, it can be deduced from the proof that $H_k(\mathcal{C}, \mathrm{Sym}^k(\mathcal{F}))$ is the S_k-coinvariants of

$$H^{-k}(L\pi_!(\mathcal{F}) \otimes^{\mathbf{L}}_B L\pi_!(\mathcal{F}) \otimes^{\mathbf{L}}_B \ldots \otimes^{\mathbf{L}}_B L\pi_!(\mathcal{F})) = H_1(\mathcal{C}, \mathcal{F})^{\otimes k}$$

Thus our claim is that this action is given by the usual action of S_k on the tensor product multiplied by the sign character. To prove this one has to work through the sign conventions in the definition of the total complex associated to a multicomplex. We omit the verification.

Lemma 2.12. *Let A be a ring. Let $P = A[E]$ be a polynomial ring. Set $I = (e; e \in E) \subset P$. The maps $Tor_i^P(A, I^{n+1}) \to Tor_i^P(A, I^n)$ are zero for all i and n.*

Proof. Denote $x_e \in P$ the variable corresponding to $e \in E$. A free resolution of A over P is given by the Koszul complex $K_\bullet$ on the x_e. Here K_i has basis given by wedges $e_1 \wedge \ldots \wedge e_i$, $e_1, \ldots, e_i \in E$ and $d(e) = x_e$. Thus $K_\bullet \otimes_P I^n = I^n K_\bullet$ computes $\mathrm{Tor}_i^P(A, I^n)$. Observe that everything is graded with $\deg(x_e) = 1$, $\deg(e) = 1$, and $\deg(a) = 0$ for $a \in A$. Suppose $\xi \in I^{n+1}K_i$ is a cocycle homogeneous of degree m. Note that $m \geq i + 1 + n$. Then $\xi = \mathrm{d}\eta$ for some $\eta \in K_{i+1}$ as $K_\bullet$ is exact in degrees > 0. (The case $i = 0$ is left to the reader.) Now $\deg(\eta) = m \geq i + 1 + n$. Hence writing η in terms of the basis we see the coordinates are in I^n. Thus ξ maps to zero in the homology of $I^n K_\bullet$ as desired. □

Proof. Let $I \subset A$ be the kernel of $A \to B$. Let $\mathcal{J} \subset \mathcal{O}$ be the kernel of $\mathcal{O} \to \underline{B}$. Then $I\mathcal{O} \subset \mathcal{J}$. Set $\mathcal{K} = \mathcal{J}/I\mathcal{O}$ and $\overline{\mathcal{O}} = \mathcal{O}/I\mathcal{O}$.

For every object $U = (P \to B)$ of $\mathcal{C}_{B/A}$ we can choose an isomorphism $P = A[E]$ such that the map $P \to B$ maps each $e \in E$ to zero. Then $J = \mathcal{J}(U) \subset P = \mathcal{O}(U)$ is equal to $J = IP + (e; e \in E)$. Moreover $\overline{\mathcal{O}}(U) = B[E]$ and $K = \mathcal{K}(U) = (e; e \in E)$ is the ideal generated by the variables in the polynomial ring $B[E]$. In particular it is clear that

$$K/K^2 \xrightarrow{\mathrm{d}} \Omega_{P/A} \otimes_P B$$

is a bijection. In other words, $\Omega = \mathcal{K}/\mathcal{K}^2$ and $\mathrm{Sym}^k_B(\Omega) = \mathcal{K}^k/\mathcal{K}^{k+1}$. Note that $\pi_!(\Omega) = \Omega_{B/A} = 0$ (Lemma 2.5) as $A \to B$ is surjective (Algebra, Lemma 00RP). By Lemma 2.10 we conclude that

$$H_i(\mathcal{C}_{B/A}, \mathcal{K}^k/\mathcal{K}^{k+1}) = H_i(\mathcal{C}_{B/A}, \mathrm{Sym}^k_{\underline{B}}(\Omega)) = 0$$

for $i < k$. This proves the final statement of the theorem.

The approach to the theorem is to note that

$$B \otimes^{\mathbf{L}}_A B = L\pi_!(\mathcal{O}) \otimes^{\mathbf{L}}_A B = L\pi_!(\mathcal{O} \otimes^{\mathbf{L}}_{\underline{A}} \underline{B}) = L\pi_!(\overline{\mathcal{O}})$$

The first equality by Lemma 2.9, the second equality by Cohomology on Sites, Lemma 08Q8, and the third equality as $\mathcal{O}$ is flat over $\underline{A}$. The sheaf $\overline{\mathcal{O}}$ has a filtration

$$\ldots \subset \mathcal{K}^3 \subset \mathcal{K}^2 \subset \mathcal{K} \subset \overline{\mathcal{O}}$$

This induces a filtration F on a complex C representing $L\pi_!(\overline{\mathcal{O}})$ with F^pC representing $L\pi_!(\mathcal{K}^p)$ (construction of C and F omitted). Consider the spectral sequence of Homology, Section 012K associated to (C, F). It has E_1-page

$$E_1^{p,q} = H_{-p-q}(\mathcal{C}_{B/A}, \mathcal{K}^p/\mathcal{K}^{p+1}) \quad \Rightarrow \quad H_{-p-q}(\mathcal{C}_{B/A}, \overline{\mathcal{O}}) = \mathrm{Tor}^A_{-p-q}(B, B)$$

and differentials $E_r^{p,q} \to E_r^{p+r,q-r+1}$. To show convergence we will show that for every k there exists a c such that $H_i(\mathcal{C}_{B/A}, \mathcal{K}^n) = 0$ for $i < k$ and $n > c$[2].

Given $k \geq 0$ set $c = k^2$. We claim that

$$H_i(\mathcal{C}_{B/A}, \mathcal{K}^{n+c}) \to H_i(\mathcal{C}_{B/A}, \mathcal{K}^n)$$

is zero for $i < k$ and all $n \geq 0$. Note that $\mathcal{K}^n/\mathcal{K}^{n+c}$ has a finite filtration whose successive quotients $\mathcal{K}^m/\mathcal{K}^{m+1}$, $n \leq m < n + c$ have $H_i(\mathcal{C}_{B/A}, \mathcal{K}^m/\mathcal{K}^{m+1}) = 0$ for $i < n$ (see above). Hence the claim implies $H_i(\mathcal{C}_{B/A}, \mathcal{K}^{n+c}) = 0$ for $i < k$ and all $n \geq k$ which is what we need to show.

Proof of the claim. Recall that for any $\mathcal{O}$-module $\mathcal{F}$ the map $\mathcal{F} \to \mathcal{F} \otimes^{\mathbf{L}}_{\mathcal{O}} B$ induces an isomorphism on applying $L\pi_!$, see Lemma 2.8. Consider the map

$$\mathcal{K}^{n+k} \otimes^{\mathbf{L}}_{\mathcal{O}} B \longrightarrow \mathcal{K}^n \otimes^{\mathbf{L}}_{\mathcal{O}} B$$

We claim that this map induces the zero map on cohomology sheaves in degrees $0, -1, \ldots, -k+1$. If this second claim holds, then the k-fold composition

$$\mathcal{K}^{n+c} \otimes^{\mathbf{L}}_{\mathcal{O}} B \longrightarrow \mathcal{K}^n \otimes^{\mathbf{L}}_{\mathcal{O}} B$$

factors through $\tau_{\leq -k}\mathcal{K}^n \otimes^{\mathbf{L}}_{\mathcal{O}} B$ hence induces zero on $H_i(\mathcal{C}_{B/A}, -) = L_i\pi_!(-)$ for $i < k$, see Derived Categories, Lemma 08Q2. By the remark above this means the same thing is true for $H_i(\mathcal{C}_{B/A}, \mathcal{K}^{n+c}) \to H_i(\mathcal{C}_{B/A}, \mathcal{K}^n)$ which proves the (first) claim.

Proof of the second claim. The statement is local, hence we may work over an object $U = (P \to B)$ as above. We have to show the maps

$$\mathrm{Tor}_i^P(B, K^{n+k}) \to \mathrm{Tor}_i^P(B, K^n)$$

are zero for $i < k$. There is a spectral sequence

$$\mathrm{Tor}_a^P(P/IP, \mathrm{Tor}_b^{P/IP}(B, K^n)) \Rightarrow \mathrm{Tor}_{a+b}^P(B, K^n),$$

see More on Algebra, Example 068F. Thus it suffices to prove the maps

$$\mathrm{Tor}_i^{P/IP}(B, K^{n+1}) \to \mathrm{Tor}_i^{P/IP}(B, K^n)$$

are zero for all i. This is Lemma 2.12. □

[2]A posteriori the "correct" vanishing $H_i(\mathcal{C}_{B/A}, \mathcal{K}^n) = 0$ for $i < n$ can be concluded.